Drug Delivery Debate: Use of Non-Viral Vectors for Nucleic Acid Vaccines



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Topics

I. What are Non-Viral Vectors?

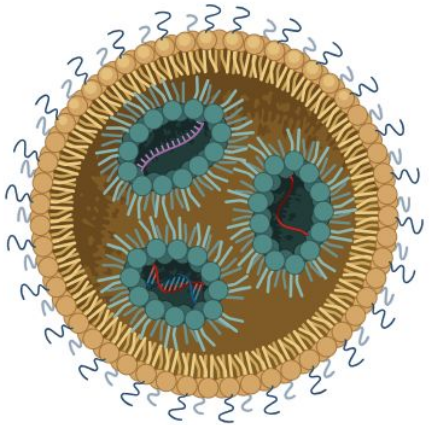
II. How do Non-Viral Vectors interact with cells and tissues?

III. Limitations of Viral Vectors

IV. Advantages of using Non-Viral Vectors

V. Examples

VI. Conclusion



What are Non-Viral Nanoparticles?

Definition: Synthetic, non-virally derived particles that can be used to **carry nucleic acids, DNA, and RNA**

1. **Lipid-Based Nanoparticles**

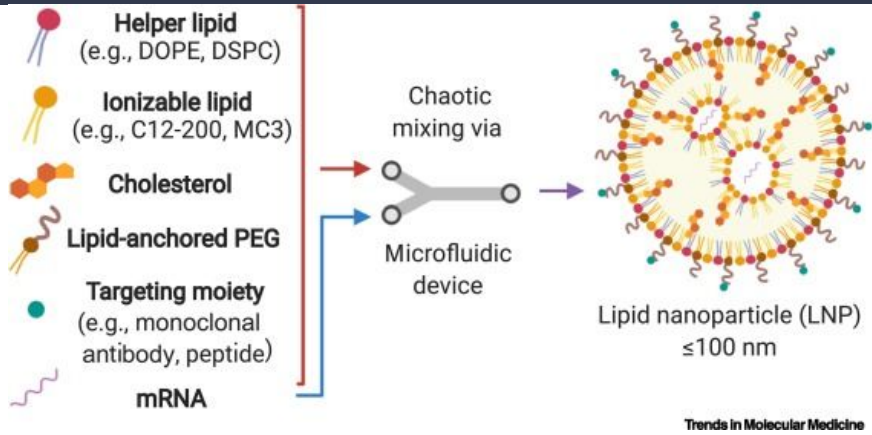
2. **Polymer-Based Nanoparticles**

3. **In-Organic Nanoparticles**

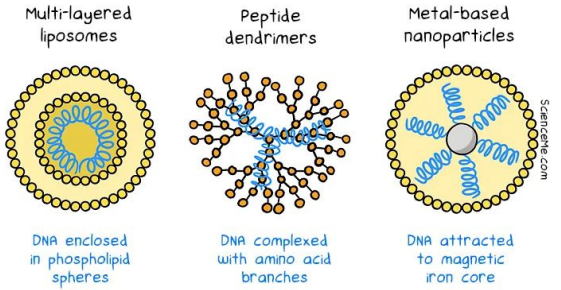
Key Qualities of Nanoparticles

Designer Nanoparticles

- Biocompatibility and low toxicity
- Tunable size → High permeability and loading capacity
  - Range of 20-500 nm in size
  - Loading capacity of 1-10 kb
- Surface modifications→ High targeting efficiency
  - Functionalized with ligands



Swingle, K. L., Hamilton, A. G., & Mitchell, M. J. (2021). Lipid Nanoparticle-Mediated Delivery of mRNA Therapeutics and Vaccines. *Trends in molecular medicine*, *27*(6), 616–617. https://doi.org/10.1016/j.molmed.2021.03.003



Mechanisms of Nanoparticle-Mediated Delivery

Protection: Shield nucleic acids from enzymatic degradation, extend blood circulation, and prevent cytotoxicity.

Cellular Uptake:

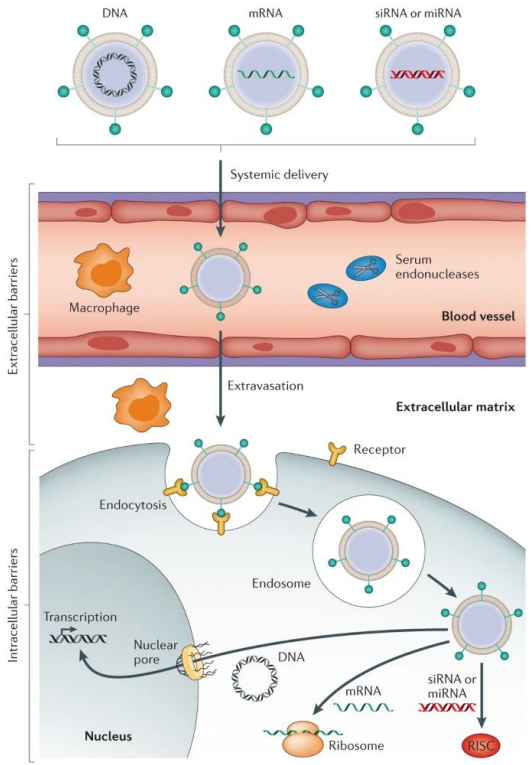
- Enter cells via phagocytosis, clathrin or caveolae-mediated endocytosis, or macropinocytosis.
- Affected by NPs' composition, size, shape, charge, surface functionalization, and hydrophobicity/hydrophilicity.

Endosomal Escape:

- NPs use ionizable lipids that protonate in acidic endosomes, disrupting membranes to release mRNA into the cytoplasm.
- Polymers like polyethyleneimine (PEI) employ the "proton sponge effect" to rupture endosomes.

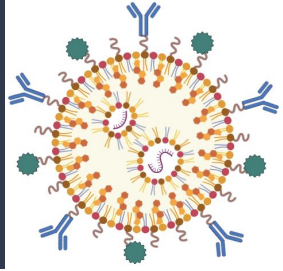
Clearance: Naturally cleared through renal, hepatic, or immune pathways.

- Can be modulated by PEGylation or "stealth" coatings



Yin, H., Kanasty, R., Eltoukhy, A. *et al.* Non-viral vectors for gene-based therapy. *Nat Rev Genet* 15, 541–555 (2014). https://doi.org/10.1038/nrg3763

Safety Benefits



- **Non-Pathogenic Background**
  - No risk of aversion to virulence or off-target effects
  - Safe for immunocompromised individuals as no replication in the body
  - Avoids latent viral genome activation concerns
- **Reduced Immune Activation → Low Immunogenicity**
  - Designed to evade immune detection, preventing premature clearance
  - Allows for multiple doses without strong anti-vector immunity
  - Lower risk of inflammatory side effects
- **Favorable Toxicity Profile → Biodegradability**
  - LNPs, polymeric nanoparticles & liposomes naturally degrade, reducing long-term toxicity.
  - Made from biocompatible materials (lipids, PLGA, PEGylated carriers)
  - Bioresorbability: no persistent presence in the body
- **Genetic & Vaccine Safety**
  - No genomic integration risk → eliminates potential for insertional mutagenesis
  - Controlled release mechanisms minimize systemic toxicity
  - Predictable dose-response curves ensure safer clinical applications

Efficacy and Stability in Preventing Viral Infections

- mRNA-based COVID-19 vaccines
  - Facilitated by LNP
    - Pfizer vaccine: 95%
    - Moderna vaccine: 94%
      - 2 doses!
- LNP Stable Composition and Function
  - phospholipids, cholesterol, (PEG)-lipids
  - mRNA vaccines utilize the host's cellular machinery to produce the target antigen
  - Mimics natural infection
- NO MUTATIONS!

Case Study: Lipid Nanoparticles and mRNA Vaccine Delivery

Targeting & Flexibility

Component optimization improves LNP stability, delivery efficiency, and reduces immunogenicity

- Mannose modified LNPs targeted antigen presenting cells improve mRNA uptake for the treatment of African swine fever virus
- Selective organ targeting (SORT) LNPs achieve tissue-specific mRNA delivery by incorporating a fifth lipid component
- Dexamethasone-loaded LNPs mitigate the immunogenicity of conventional LNP delivery systems

Surface modifications enhance LNP functionality, enabling targeted delivery and improved cell membrane penetration.

- Antibodies: PECAM-1 antibody-conjugated LNPs enable lung-targeted delivery
- Peptides: RVG29-functionalized LNPs facilitate blood-brain barrier crossing

Zong, Yan et al. Advanced materials vol. 35,51 (2023)

Advantages over Viral Vectors

- **Versatility & Customizability**
  - Compatible with mRNA, siRNA, DNA, CRISPR, etc.
  - Engineered for targeted delivery to specific cells
  - Tunable release mechanisms (pH-sensitive, enzyme-sensitive, etc)
- **Scalability & Cost-Effective Manufacturing**
  - Synthetically produced → rapid scale large production
  - Easier storage & transport
  - Lower regulatory hurdles due to no viral
- **Repeat Dosing, Stability, and Flexibility**
  - Can be administered multiple times w/o triggering anti-vector immunity
  - Boosters, variant updates, personalized medicine
  - Protective encapsulation prevents premature degradation of nucleic acids
  - Compatible with lyophilization, improving shelf-life

	Viral Vectors	Non-viral Vectors
Origin	Derived from viruses	Synthetic
Transfection efficiency	High	Low (Higher with LNPs)
Immune reactions	More likely	Less likely
Integration of DNA	More likely	Less likely
Packaging capacity	max 5-8 kb (no mRNA possible)	22 kb mRNA possible
Applications	Gene therapy, gene editing	Gene therapy, gene editing, gene silencing, drug delivery

Overcoming Viral Vector Limitations I

Features	Viral Vector Limitations	NPs
Immunogenicity	High immunogenicity; pre-existing neutralizing antibodies prevent redosing	Low immunogenicity can be redosed; component optimization leads to reduced immunogenicity
Genomic Integration Risk	Lentivirus & retrovirus may integrate into the genome, causing insertional mutagenesis	No genomic integration; transient expression reduces oncogenic risks

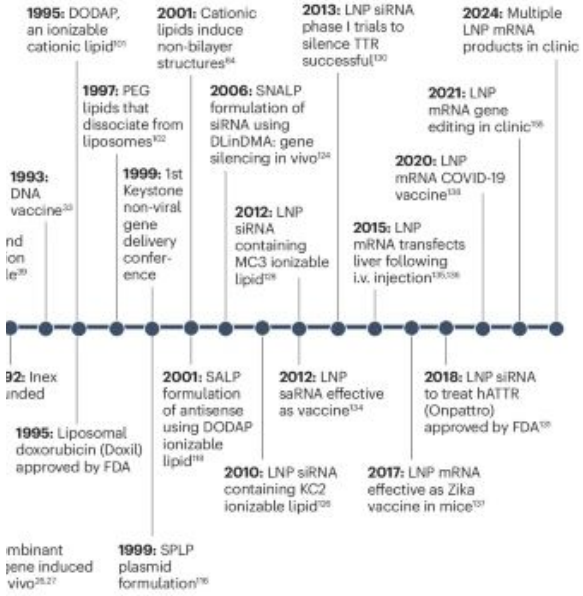
Shirley, Jamie L. et al. Molecular Therapy, Volume 28, Issue 3, 709 - 722

Overcoming Viral Vector Limitations II

Features	Viral Vector Limitations	NPs
Cargo Capacity	Limited to ~4.7 kb of DNA/RNA	Can encapsulate large nucleic acids (mRNA, siRNA, CRISPR, etc.) >10 kb
Manufacturing & Scalability	Requires mammalian or insect cell production, making it expensive and time-consuming	Synthetic, scalable production using synthetic formulation, making it cost-effective

Non-Viral Vector Examples

- **Stabilized Nucleic Acid Lipid Particles (SNALPs)**
  - Reduce Hepatitis B Virus using siRNA
- **LNP:**
  - mRNA vaccines
    - COVID-19: **13.5 billion COVID-19 vaccine doses**
- **PLGA: poly(lactic-co-glycolic acid) nanoparticles**
  - Inactivated Vaccines
    - Inactivated Polio Vaccine
    - Influenza vaccine
- **Gold nanoparticles:**
  - Conjugate Vaccines
    - Pneumococcal Vaccines: protect against *Streptococcus pneumoniae* infections
- **Alum-based nanoparticles:**
  - Toxoid Vaccines
    - Tetanus Vaccine



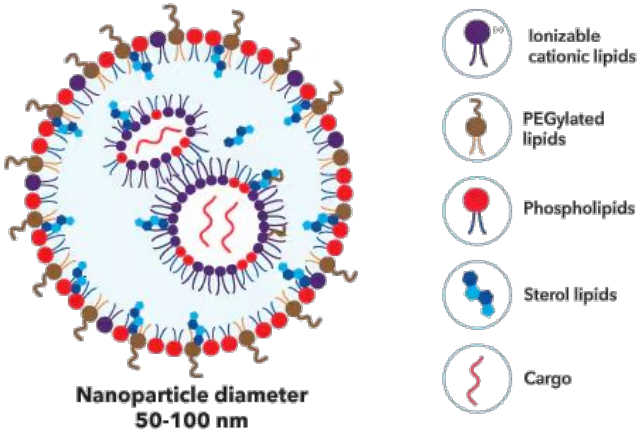
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[Lipid nanoparticle–based COVID-19 vaccines: Ensuring pharmaceutical stability, safety, and efficacy | American Journal of Health-System Pharmacy | Oxford Academic]

Conclusion

Non-Viral Nanoparticles are **more effective** at delivering nucleic acid vaccines for viral infection prevention.

1. **Tunable Characteristics**
   a. Cargo Capacity
   b. Targeting Capabilities
2. **Safer than Viral Vectors**
   a. Lower Risk of Adverse Immune Effects
   b. No Genomic Integration
3. **Better Scalability to Clinic**



Thank you! Questions?

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